Summary

CSC3034 Computational Intelligence

Dr Tang Tiong Yew

Lecture outline

Summary

Evolutionary Computation

Artificial Neural Networks

Fuzzy Systems

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Evolutionary Computation

- Genetic Algorithm
- Particle Swarm Optimisation
- Ant Colony Optimisation

Artificial Neural Networks

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Particle Swarm Optimisation

Artificial Neural Networks

Supervised Learning

Unsupervised Learning

Fuzzy Systems

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    - Feedforward
    - Recurrent
  - Unsupervised Learning
- Fuzzy Systems

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    - Kohonen
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  - Fuzzy sets operations
  - Hedges
  - Inference Methods

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    - Mamdani
    - Sugeno

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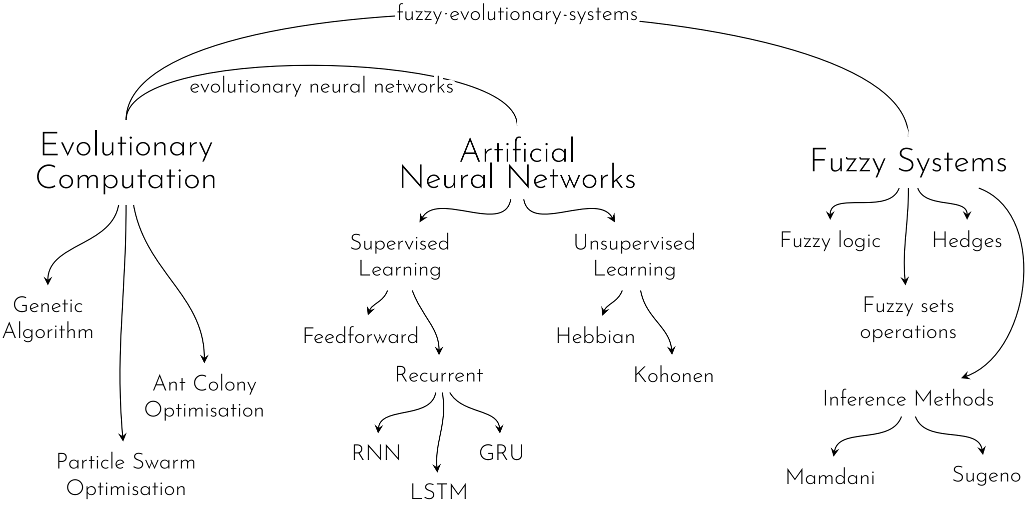

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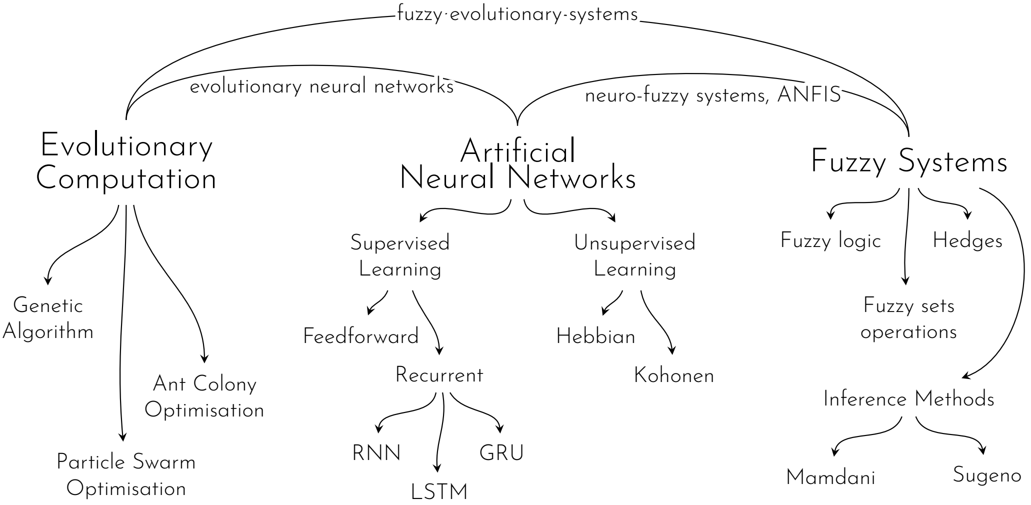
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Performance metrics

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Errors

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Mann-Whitney
U test

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Contingency matrix

Summary

Confusion matrix

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ROC curves

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Final Exam Format

- 3 compulsory questions ($2 \times 30\% + 1 \times 40\% = 100\%$)

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  - Hybrid system problem related to the earlier part of the question
- Q3
  - Problem related to performance metrics

Final Exam Scope

1. **Evolutionary Computation**

   Genetic algorithm · Particle swarm optimisation

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   Perceptron · Feedforward multilayer neural networks

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   Evolutionary neural network · Fuzzy evolutionary system

4. **Performance Metrics**

   Confusion matrix · Contingency matrix · Accuracy · Precision · Recall · Specificity · Mann-Whitney U test · Chi-square test

Logistics

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- Late arrival is only permitted up to 9:30 a.m.

Sample Questions for Final

a) A feedforward neural network with 1 hidden layer of 3 neurons was designed to learn the decision making process for the prediction of marks obtained by a student based on the average number of hours in a day the student spent studying and the interest level of the student. All inputs and outputs are scaled to the range of 0 to 1. Sketch the designed neural network. [5 marks]

b) The neural network in (a) is initialised such that all the weights have the value of 0.12 and all the thresholds of the neurons are 0.05. In the first iteration, given a data point where the average number of hours in a day the student spent studying is 0.2, the interest level is 0.7, and the marks obtained is 0.75, calculate (i) the new value for the weight of the connection from the first input neuron to the first hidden neuron and (ii) the new value for the threshold of the first hidden neuron after the iteration given that the learning rate α is 0.3. [15 marks]

c) You are tasked to identify the optimal number of hidden layers and hidden neurons to obtain the best performance for the neural network in (a). What is the Computational Intelligence (CI) approach suitable for this purpose? Briefly explain the mechanism of the hybrid system. [10 marks]

Sample Questions for Final

A client handed metal of volume V_T over to a metalsmith to be melted and made into metal balls. By using Equation (9) as the fitness function to be minimised, particle swarm optimisation (PSO) is used to identify the optimal radius of each metal ball. fitness function, $f(r) = \frac{4}{3}\pi r^3 + \frac{3V_T}{r}$

a) Define three appropriate termination conditions for a PSO algorithm. [3 marks]

b) Describe the difference between global best PSO and local best PSO from the aspect of the mechanism. Discuss the effect of the difference. [6 marks]

Continuation from last slide,

c) Assuming the local best PSO is used to solve the optimisation problem. The neighbourhood is defined by the particle indices. Consider each neighbourhood consists of 3 particles. At initialisation, the particles are initialised as in Table A. Identify the neighbourhood best position of each particle. Use $V_T = 10$. [5 marks]

Table A: Initialised particles

Particle	Radius of each metal ball, r
p_1	0.5
p_2	3.0
p_3	2.7
p_4	0.3
p_5	1.5
p_6	1.0
p_7	4.5
p_8	6.9
p_9	0.2
p_{10}	8.7

Continuation from last slide,

d) Global best PSO and local best PSO are used to solve the optimisation problem. Each algorithm is executed 5 times and the best positions are recorded in Table B. Using Mann-Whitney U test, determine if one algorithm performs better than the other algorithm. [11 marks]

Table B: Result of global best PSO and local best PSO

Run	Result of global best PSO	Result of local best PSO
R_1	1.25	1.24
R_2	1.15	1.22
R_3	1.30	1.14
R_4	1.32	1.16
R_5	1.23	1.20

Sample Questions for Final

a) A Mamdani-style fuzzy inference system to predict traffic situation based on the time of day and the temperature is to be designed. [20 marks]

Design and sketch the fit vectors for the linguistic variables specified with their respective linguistic values (fuzzy sets) and numerical ranges in Table X.

b) Over the years, Rachel has collected the data of the traffic situation with the corresponding temperature and time of day. An expert has provided 5 fuzzy rules to support the decision making. Suggest a hybrid system for Rachel to validate the 5 fuzzy rules based on the collected data. Briefly explain how this system achieves the validation. [10 marks]

Table X: Specification of linguistic variables

Linguistic value	Numerical range
Variable: Time of day	
Morning	00:00 - 12:00
Noon	11:00 - 13:00
Afternoon	12:00 - 17:00
Evening	16:00 - 19:30
Night	19:00 - 23:59
Variable: Temperature	
Cold	$< 25^\circ C$
Warm	$20 - 35^\circ C$
Hot	$> 33^\circ C$
Variable: Traffic	
Free-flowing	< 80 cars per km
Congested	> 70 cars per km

Sample Questions for Final

a) Kayden has trained an artificial neural network to predict a student's overall performance as one of the three classes, Excellent, Average, Poor, based on the score of the student in Quiz 1.

The trained neural network is used to predict the performance of 35 students. Out of the 10 students who are predicted to be Excellent, 4 obtained Excellent, 3 Average, and 3 Poor. Of the 15 students predicted to be Average, 10 obtained Average and 5 obtained Excellent. Among the 10 students predicted to obtain Poor, 9 obtained Poor and 1 Average.

i) Construct the confusion matrix of the prediction. [9 marks]

ii) Calculate the class confusion matrix of the prediction. [7 marks]

iii) Discuss the performance of the neural network in predicting a student's overall performance based on the student's score in Quiz 1. [4 marks]

Continuation from last slide,

b) A lecturer used the trained neural network in (a) to predict the performance of students from two different subjects. For each subject, the number of students predicted to obtain each grade and the number of students actually obtain the grade are specified in Table Y and Z. [20 marks]

Table Y:
Number of students for each grade in subject $\mathbb{A}$

		Number of students	
		Predicted	Actual
Grade	Excellent	10	5
	Average	20	23
	Poor	5	7

Table Z:
Number of students for each grade in subject $\mathbb{B}$

		Number of students	
		Predicted	Actual
Grade	Excellent	25	40
	Average	30	20
	Poor	40	35

Using χ^2 test, at 0.05 level of significance, evaluate if the neural network performs better in predicting the grades of the students in subject $\mathbb{A}$ or subject $\mathbb{B}$.

CSC3034 Current Course Update

Summary - presentation

Synchronized from the current CSC3034 course site on 14 August 2026.

- Current labs connect core CI algorithms with physical-AI simulation examples.
- Isaac Sim examples use its bundled Python 3.12 environment and current launchers.
- Every simulator-dependent example includes a CPU/Matplotlib learning fallback.

Current materials author: Dr Tang Tiong Yew

See the synchronized lab notes and runnable examples for full instructions.